AT82.05 Artificial Intelligence: Natural Language Understanding

Multimodal and Agentic RAG for Grade 10 Mathematics Tutoring in the Nepalese Curriculum

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PROBLEM STATEMENT

- Secondary school students in Nepal, especially in rural areas, struggle with Grade 10 mathematics
- Limited access to qualified teachers and structured academic support
- Students rely on memorization rather than conceptual understanding
- Existing AI tools generate generic, unstructured responses not aligned with Nepal's curriculum
- Most tutoring systems only support text input, not real-world handwritten or printed problems

RESEARCH QUESTIONS

- **RQ1 (RAG Effectiveness)**: Does a curriculum-aligned Retrieval-Augmented Generation system improve the correctness and quality of mathematical solutions compared to baseline LLMs?

- **RQ2 (Multimodal Impact)**: How does the inclusion of visual information, such as diagrams, affect the quality and interpretability of mathematical problem-solving?

- **RQ3 (Agentic Workflows)**: Do agentic RAG workflows improve solution correctness and reliability compared to static RAG pipelines?

PROPOSED SOLUTION

- This project proposes an advanced RAG-based math tutoring system with multimodal retrieval and agentic workflows
- Supports text and image inputs via OCR for unified retrieval
- Extracts and converts textbook diagrams into structured representations
- Combines text and image embeddings for joint retrieval
- Uses Self-RAG, CRAG, and Adaptive RAG via LangGraph for smart retrieval
- Dynamically evaluates and refines outputs for better accuracy

RELATED WORKS

AREA	KEY WORK	INSIGHT
LLMs for Math	Cobbe et al. (2021), Wei et al. (2022)	Chain-of-thought improves reasoning but lacks curriculum grounding
RAG	Lewis et al. (2020), Karpukhin et al. (2020)	Retrieval grounds output in external knowledge
Retrieval Design	DesignDPR (Karpukhin et al., 2020)	Dense retrieval outperforms sparse methods
Multimodal / OCR	Tesseract (Smith, 2007)	Lightweight OCR enables image-based math input

RESEARCH GAP

- LLMs rely on internal knowledge and lack curriculum alignment
- Existing RAG systems are built for general tasks, not education
- Multimodal models are not designed for math diagrams
- Agentic RAG has never been applied to math tutoring
- No system combines all three — retrieval, vision, and agentic reasoning

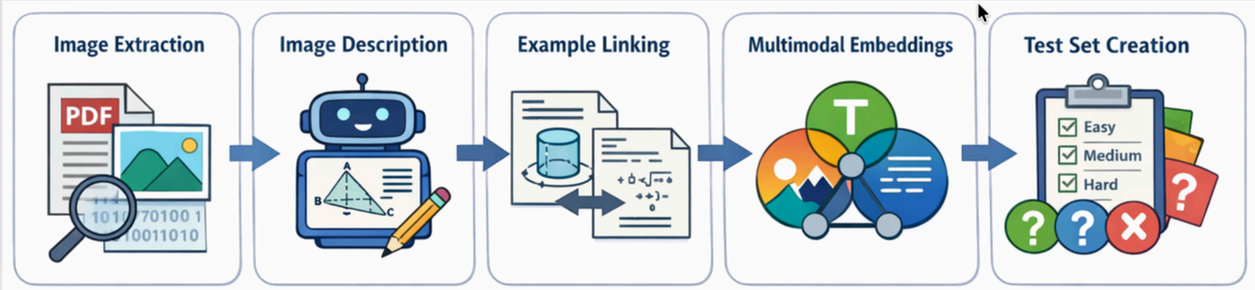
DATASET

Dataset:

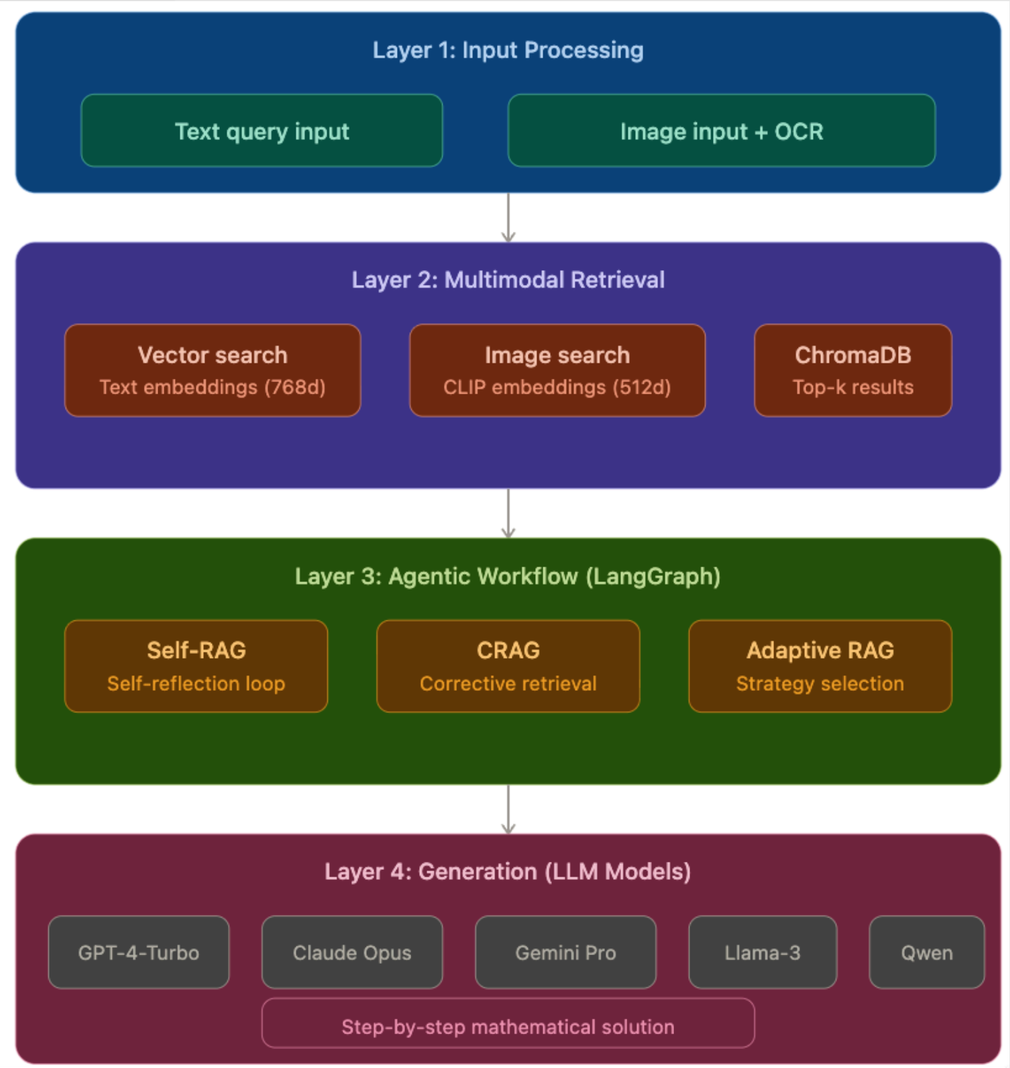
- Source: Official Nepal Grade 10 Mathematics Textbook (PDF)
- Three data types: full chapter text, worked examples, exercise questions
- Worked examples are the primary knowledge base, with metadata (chapter, page, example index)
- Stored in JSON format, indexed into a ChromaDB vector database

DATA PREPROCESSING

	Stage	Tool/Method	Output
1	Image Extraction	PyMuPDF	Images are extracted with metadata (page, position, dimensions)
2	Image Description	GPT-4V	Each image is described with type, labels, measurements, and math concepts
3	Image-Example Linking	Page proximity + semantic similarity	Diagrams are paired with their most relevant worked examples
4	Multimodal Embeddings	BERT (768d) + CLIP (512d) + Desc (768d)	Three embeddings are combined into 2048d vectors and indexed in ChromaDB
5	Test Set Creation	Stratified sampling	100 questions are labeled by difficulty, type, and visual requirement



SYSTEM ARCHITECTURE



EXPERIMENT DESIGN

Experiment 1 — Baseline LLM vs RAG: Does RAG improve math problem-solving over baseline LLMs?

Model	Setting	BLEU	ROUGE-L	BERT
GPT-4-Turbo	Baseline	-	-	-
GPT-4-Turbo	RAG	-	-	-
GPT-3.5-Turbo	Baseline	-	-	-
GPT-3.5-Turbo	RAG	-	-	-
Claude Opus 4.6	Baseline	-	-	-
Claude Opus 4.6	RAG	-	-	-
Qwen2-Math-7B	Baseline	-	-	-
Qwen2-Math-7B	RAG	-	-	-
Gemini 1.5 Pro	Baseline	-	-	-
Gemini 1.5 Pro	RAG	-	-	-

Table 1: Baseline LLM vs RAG comparison.

EXPERIMENT DESIGN

Experiment 2 — Multimodal RAG: Does adding visual diagrams improve geometry problem-solving?

Model	Modality	BLEU	ROUGE-L	BERT	Time
GPT-4-Turbo	Text-only	-	-	-	-
GPT-4-Turbo	Text+Desc	-	-	-	-
GPT-4V	Text+Image	-	-	-	-
Claude Opus	Text+Image	-	-	-	-
Gemini Vision	Text+Image	-	-	-	-
Qwen2-Math	Text-only	-	-	-	-

Table 2: Multimodal RAG comparison.

EXPERIMENT DESIGN

Experiment 3 — Agentic RAG: Do Self-RAG and CRAG improve correctness over static RAG?

Model	Workflow	BLEU	ROUGE-L	BERT	Time
GPT-4-Turbo	Baseline	-	-	-	-
GPT-4-Turbo	Self-RAG	-	-	-	-
Claude Opus	Self-RAG	-	-	-	-
GPT-4-Turbo	CRAG	-	-	-	-
Claude Opus	CRAG	-	-	-	-

Table 3: Agentic RAG comparison.

EXPECTED OUTCOMES

- RAG is expected to improve correctness and structure of solutions compared to baseline LLMs
- Hybrid retrieval (semantic + keyword) is expected to retrieve more relevant context than vector-only methods
- Multimodal integration is expected to improve performance on geometry and visual problems, but increases cost and latency
- Visual inputs are expected to have limited impact on non-visual problems such as algebra and arithmetic
- Self-RAG and CRAG are expected to improve correctness through iterative refinement, at the cost of additional latency

THANK YOU

Do you have any questions?